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Top Skills

Forging
Annealing
Welding

Languages

English (Native or Bilingual)
Sicilian (Limited Working)

Honors-Awards

Rensselaer Leadership Award
Women's Leadership Scholarship
Lila & Gerald Ellsworth'50
Scholarship
TMS Bladesmithing Competition:
Special Citation for Multi-Functional
Design

Cristina Ninfa Gandolfo

Incoming ChemE Grad Student @ NYU | B.S. Material Science and
Engineering RPI '26
New York City Metropolitan Area

Summary

Materials Science and Engineering graduate from Rensselaer Polytechnic Institute and incoming M.S. student in Chemical Engineering at New York University with interests spanning advanced materials, polymers, sustainable technologies, and innovation-driven product development. My experience combines technical research, materials characterization, project leadership, and organizational management. As President of RPI's Material Advantage chapter, I led initiatives focused on professional development, industry engagement, and student growth while fostering collaboration across multidisciplinary teams. I am particularly interested in translating scientific discovery into practical solutions, whether through research, manufacturing, product development, or entrepreneurial ventures. My approach to engineering is inherently interdisciplinary, drawing from technical expertise, business acumen, and leadership experience to tackle complex challenges and create meaningful impact. I am always interested in connecting with researchers, engineers, entrepreneurs, and industry leaders working at the forefront of innovation.

Experience

Rensselaer Polytechnic Institute
3 years 10 months

Work Study Student
August 2022 - May 2026 (3 years 10 months)
Troy, NY

- Managed front-office operations for the Lally School of Business Student Services suite, serving as the primary point of contact for students and visitors.
- Streamlined academic advising logistics by managing advisor calendars, scheduling student appointments, and maintaining confidential records with strict adherence to privacy protocols.

- Provided comprehensive administrative support, including handling phone correspondence, managing document filing, preparing materials for advisor sessions, and coordinating logistical setup for school events and professional development workshops.

Undergraduate Researcher

January 2025 - April 2025 (4 months)

Troy, NY

- Developing method to use Spent Coffee Grounds in PHA/PLA as an alternative to synthetic plastics for applications in acoustic paneling.

Undergraduate Researcher

January 2024 - June 2024 (6 months)

Troy, NY

- Conducted a comprehensive literature review on commercial bio-composites to inform the design of new experimental methods aimed at investigating fiber architecture effects on mechanical properties.
- Overcame critical processing challenges by developing and incorporating a fold-mixing method, which significantly reduced air bubble defects in the cured composite matrix.
- Established a repeatable fabrication and testing protocol to evaluate tensile and flexural performance under challenging loading conditions.

Resident Assistant

August 2023 - May 2024 (10 months)

Troy, New York, United States

- Mentored a community of freshman residents by facilitating one-on-one meetings, mediating conflicts, and connecting students with RPI academic and wellness resources.
- Managed crisis situations and upheld safety standards through nightly duty shifts, regular facility inspections, and the immediate documentation of Incident Reports (IR).
- Served as a liaison between residents and the Office of Student Living and Learning, ensuring clear communication of campus policies and health updates.

STEP/PREFRACE Lead Outreach Coordinator

June 2023 - August 2023 (3 months)

Troy, NY

- Led and managed lab budget, coordinated team of assistants, and designed lesson plans on laboratory safety and engineering ethics.

- Designed and delivered interactive hands-on lab modules focused on polymers, degradation, and fundamentals in material science and engineering.
- Mentored students individually on research and college pathways, increasing post-program interest in RPI's MSE department by five students.

Undergraduate Researcher

January 2023 - May 2023 (5 months)

- Investigated the galvanic and electrolytic growth of copper and tin dendrites via electrochemical growth using copper sulfide and tin chloride.
- Refined the experimental design and developed a lab procedure that was later integrated into RPI's Introduction to Materials Science course where it remains in use three years later.

GlobalFoundries

Advanced Lithography Process Engineering Intern

May 2025 - August 2025 (4 months)

Malta, New York, United States

- Developed a metrology-driven strategy to scan test wafers for critical dimension (CD) measurements, enabling earlier fault detection in production wafers.
- Integrated CD measuring on various chem stacks, layers and lithography tracks, addressing potential equipment and resist batch qual issues.
- Optimized sampling method on the equipment monitor to reduce lithography tool downtime, improving data quality while preserving the standards of measurement on production.

SABIC

Research And Development Co-op

June 2024 - December 2024 (7 months)

Pittsfield, Massachusetts, United States

- Designed and executed a study to evaluate the suitability of carbon fiber-reinforced thermoplastic resin for use in Big Area Additive Manufacturing printed aerospace components.
- Performed optical microscopy, CT scan imaging, mechanical testing, and data analysis via JMP software to correlate optimal print path and fiber orientation with tensile performance.
- Designed and implemented a test protocol to evaluate surface adhesion and coating behavior of thermoplastics in automotive applications.
- Recommended new tooling specifications (mirror-polished

cavities) to improve repeatability and surface quality across injection-molded samples.

- Conducted accelerated weathering and mechanical testing on resin formulations to assess UV stability, tensile retention, and microstructural integrity over time.

- Collaborated with formulation chemists and product developers to inform resin development for harsh-environment performance.

Education

Rensselaer Polytechnic Institute

Bachelor of Science, Materials Science and Engineering, Minor in Economics · (August 2022 - May 2026)